PRODUCT REQUIREMENTS DOCUMENT

VOUCH

On-Chain Reputation Infrastructure for the Agent Economy

A Ziki Labs Product

Version	1.0.0	Final Draft
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Target Chain	Base L2 (Ethereum Layer 2)
Classification	CONFIDENTIAL	INTERNAL USE ONLY
Prepared By	Ziki Labs	Engineering Leadership

1. Executive Overview

1.1 About Ziki Labs

Ziki Labs is a web3 product studio and builder community operating as a mother company for internally developed decentralized infrastructure products. Ziki Labs identifies structural gaps in emerging technology markets, assembles focused engineering teams, and ships protocol-grade products designed to become foundational primitives in their respective ecosystems. VOUCH is the first flagship product to emerge from the Ziki Labs portfolio.

1.2 Product Vision

VOUCH is the trust layer for the AI agent economy. It is a decentralized reputation infrastructure built on Base L2 that solves the most critical unsolved problem in autonomous agent deployment: there is no credible, tamper-resistant, sybil-resistant mechanism to verify whether an agent is actually good at its job.

As autonomous agents increasingly hire humans, execute tasks, write code, manage finances, and transact on-chain, every interaction begins from zero trust. Every operator starts with nothing. Every task poster takes a blind bet. VOUCH ends that.

The core primitive of VOUCH is a non-transferable ERC-721 Reputation Soulbound Token (rSBT) minted per agent, whose on-chain score evolves with every verifiably completed task. Over time, an agent's VOUCH score becomes its professional track record: portable, permanent, and provably honest.

> **One-Line Value Proposition**
>
> VOUCH is an agent DAO that manages and monetizes AI agent reputation for Locus Tasks, enabling trustless hiring and payment based on verifiable on-chain performance history.

1.3 Why This, Why Now

- The AI agent economy is crossing the line from demos to production. Agents are autonomously booking meetings, writing code, executing trades, and managing workflows at scale.
- No credible on-chain identity or reputation layer exists specifically for AI agents. Every deployment starts with zero provenance and zero accountability.
- Locus Tasks and similar agentic task marketplaces need a reputation oracle to differentiate high-quality agents from low-quality or malicious ones. Without it, the marketplace cannot scale.
- Regulatory pressure from the EU AI Act and the US Executive Order on AI is pushing for auditability and accountability mechanisms. VOUCH provides both by design, not as an afterthought.

- The market timing for 'Know Your Agent (KYA)' as a B2B infrastructure primitive is optimal: early enough to own the category, late enough that demand is fully validated.
- Ziki Labs is positioned to build this because it operates at the intersection of web3 infrastructure, AI agent tooling, and community-driven builder culture, all of which are required to get this right.

1.4 Confirmed Blockchain Architecture

Primary Chain	Base L2 (Ethereum L2, OP Stack) - EVM-compatible, low gas fees, Coinbase-backed ecosystem
Fallback / Bridge	Ethereum Mainnet for governance-critical votes and final state anchoring
Payment Token	USDC (native Circle CCTP on Base) - stable, widely integrated, regulatory-friendly
Reputation Token	ERC-721 non-transferable Soulbound Token (rSBT) - one per agent address
Governance Token	ERC-20 $VOUCH - earned by contributors; powers DAO voting via Governor Bravo fork
Oracle	Chainlink Functions for off-chain task-completion verification callbacks
Storage	IPFS and Filecoin for audit trails; on-chain stores content hashes only
Indexer	The Graph Protocol subgraph for real-time event indexing and GraphQL queries

2. Market Analysis and User Personas

2.1 Market Sizing

The agentic AI market is not a niche. It is the next operating layer of the global economy. VOUCH is positioned as infrastructure within that layer, not an application on top of it.

Segment	Size	Basis
TAM (Total Addressable Market)	$52B by 2030	Global agentic AI market at current CAGR of 46.3% (Gartner, MLM 2026)
SAM (Serviceable Addressable Market)	$6.8B	Task-marketplace and AI hiring platforms with on-chain or API settlement capability
SOM Year 1	$12M ARR	3,000 active agents at 200 tasks/month, 1.5% fee on average $20 task value
SOM Year 3	$85M ARR	Ecosystem expansion across 5+ task marketplaces; KYA API licensing

2.2 Market Tailwinds Specific to VOUCH

- Gartner predicts over 40% of enterprise applications will embed task-specific AI agents by end of 2026, creating massive demand for agent credentialing.
- 93% of IT leaders report plans to introduce autonomous agents within two years, yet governance and accountability tooling is almost entirely absent from current stacks.
- Multi-agent architectures are becoming the standard design pattern (1,445% surge in multi-agent system inquiries, Q1 2024 to Q2 2025 per Gartner), which amplifies the need for per-agent identity.
- The Anthropic Model Context Protocol (MCP) and Google A2A are establishing interoperability standards for agents, which will accelerate cross-platform adoption of VOUCH as a shared reputation layer.
- Regulatory tailwinds: EU AI Act high-risk AI system logging requirements and US AI accountability frameworks create compliance-driven demand for VOUCH's immutable audit trail.

2.3 User Personas

Persona A: The Autonomous Agent (Non-Human System Actor)

The primary system actor in the VOUCH protocol. An AI agent (LLM-based, rule-based, or hybrid) deployed by an operator. The agent holds an EOA wallet, carries its rSBT, bids on tasks, receives USDC payments, and accumulates on-chain reputation through verifiable task completion. It interacts with VOUCH smart contracts via its operator's execution environment.

- **Core needs:** A portable, tamper-proof track record. Premium task access at high reputation tiers. Low-friction staking and payment escrow. Cross-platform portability so reputation does not reset on every new marketplace.
- **Primary pain point:** Starts every deployment with zero trust and zero history. Task posters must evaluate agents subjectively or via centralized ratings that are trivially gamed.

Persona B: The Task Poster (Human or Organization)

A human professional, startup, or enterprise posting tasks on Locus Tasks or any integrated marketplace. Wants provably capable agents, not black-box unknowns. Willing to pay a meaningful premium for high-reputation agents because the cost of a failed task exceeds the premium many times over.

- **Core needs:** Transparent, objective agent scoring. Escrow protection that guarantees payment only upon verified completion. Full audit trail for compliance and accountability.
- **Primary pain point:** No reliable way to distinguish battle-tested agents from freshly deployed, low-quality, or malicious actors. Every hire is a gamble.

Persona C: The Agent Operator (Developer or Organization)

The human or team that built, maintains, and deploys the agent. Wants their agent to earn reputation capital that compounds over time into higher earnings and greater access. Will stake $VOUCH tokens to upgrade agent tiers. Operates one or many agents simultaneously.

- **Core needs:** Fleet dashboard with visibility across all agents. Staking and tier management interface. Formal dispute mechanism when reputation scores are unfairly affected.
- **Primary pain point:** No portable, cross-platform reputation that survives marketplace churn. An agent with a great track record on one platform starts from zero on the next.

Persona D: The DAO Contributor (Governance Participant)

Token holders who govern the VOUCH protocol. This group includes early investors, active operators with large $VOUCH holdings, and community builders who contribute to protocol development. They vote on scoring algorithm parameters, fee structures, dispute outcomes, treasury allocation, and protocol upgrades.

- **Core needs:** A clean governance interface. Transparent treasury reporting with real-time data. Voting power that is meaningful and protected from whale capture.
- **Primary pain point:** DAO governance is historically low-participation, easily captured, and technically inaccessible to non-developers. VOUCH must design against this from day one.

Persona E: The Enterprise Integrator (Platform or Protocol)

A third-party task marketplace, AI platform, or enterprise workflow system that wants to integrate VOUCH's KYA standard into their own product. They consume the VOUCH API to surface reputation scores, verify agent identity, and access audit trails without building this infrastructure themselves.

- **Core needs:** A clean REST and GraphQL API. Documented webhook integration. SLA guarantees. Pricing that makes sense for high-volume programmatic queries.
- **Primary pain point:** Building their own agent credentialing system requires months of engineering effort with no ecosystem network effects. VOUCH provides it as a shared layer.

3. Problem Statement

The AI agent economy is scaling faster than the trust infrastructure that needs to support it. The following table maps each structural problem to its current workaround and why that workaround fails at scale.

Problem	Current Workaround	Why It Fails at Scale
No portable agent reputation	Centralized platform ratings	Siloed per-marketplace, easily gamed, and lost entirely when operators switch platforms
No sybil resistance for agents	Email or API-key-based accounts	Trivially duplicated with near-zero cost; adversarial operators spin up thousands of fake agents
No on-chain performance proof	Self-reported metrics	Completely unverifiable; subject to trivial manipulation; trusted by no serious task poster
No escrow for agent payments	Trust-based invoicing or manual settlement	High chargeback risk and zero delivery guarantee; chills adoption on both sides of the market
No collective governance for agent standards	Platform-unilateral policy decisions	Centralized; no operator recourse; susceptible to arbitrary rule changes and regulatory capture
No KYA (Know Your Agent) standard	Nothing - the gap is open	Regulatory exposure for enterprises deploying agents; safety blindspot for the entire ecosystem
No cross-platform reputation portability	Starting from scratch on each platform	Destroys the incentive for operators to invest in agent quality; no long-term flywheel

Root Cause

Agent identity is ephemeral, platform-local, and non-verifiable. Without a decentralized reputation primitive that follows agents across platforms, the agent economy cannot scale trustlessly. Every interaction continues to require costly centralized intermediation. VOUCH is the primitive.

4. Success Metrics and OKRs

4.1 MVP Metrics (Months 0 to 6)

Metric	Target	How Measured
Agents Onboarded	500 unique agent addresses with minted rSBT	On-chain AgentRegistry event logs
Tasks Completed via Escrow	2,000 tasks with Chainlink-verified completion	TaskEscrow contract finalization events
Reputation Score Update Latency	Less than 5 seconds from oracle callback to score write	Chainlink Functions callback timing logs
Smart Contract Audit Outcome	Zero critical findings; all high findings resolved	Published third-party audit reports
Frontend Uptime	99.5% monthly availability	Uptime monitoring dashboard (Grafana)
Gas Cost Per Task Lifecycle	Less than $0.10 total on Base L2	On-chain gas tracker and cost model
Task Poster Conversion	30% of registered posters complete their first escrowed task	Funnel analytics
Time to First Task (Agent)	Less than 24 hours from registration to first accepted task	Event log timestamps

4.2 Growth Metrics (Months 6 to 18)

Metric	Target	How Measured
Monthly Active Agents	3,000 MAA performing at least 1 task per month	Wallet activity and task event analytics
DAO Treasury Balance	$500K USDC minimum	Treasury contract balance
Protocol Fee Revenue	$12K MRR by month 12	Fee collector contract
Governance Participation	Greater than 15% quorum on all major proposals	Governor contract vote records
Dispute Resolution Time	Less than 72 hours median resolution	DisputeManager contract timestamps
Task Poster 90-Day Retention	Greater than 65%	Locus Tasks integration analytics

Metric	Target	How Measured
Agent Tier Upgrade Rate	30% of agents advance at least one tier within 90 days	rSBT tier attribute evolution events
KYA API Third-Party Integrations	At least 3 external platforms consuming VOUCH API by month 18	API key issuances and usage logs

5. Product Scope

MVP Boundary Principle

The MVP proves the core loop: Agent registers and receives an rSBT. Agent accepts a task. USDC is escrowed. Work is completed off-chain. Chainlink verifies completion. Reputation score updates on-chain. USDC releases to agent. Everything outside this loop is Phase 2 or later.

5.1 Phase 1: MVP (Months 0 to 4)

5.1.1 Smart Contracts

- AgentRegistry.sol: Agent registration, rSBT minting (ERC-721 non-transferable), tier management, and operator-to-agent wallet mapping
- ReputationEngine.sol: Deterministic scoring algorithm, score read and write, tier threshold enforcement, EWMA dampening
- TaskEscrow.sol: USDC escrow, full task lifecycle state machine, Chainlink verification trigger, payment release on completion
- DisputeManager.sol (basic): Two-party dispute flagging with founding committee multisig resolution and on-chain outcome recording
- DAOGovernor.sol: Governor Bravo fork with $VOUCH token voting, proposal lifecycle, quorum enforcement, and timelock integration
- VOUCHToken.sol: ERC-20 governance token with vesting support, staking interface, and delegation capability

5.1.2 Backend Services

- Verification Oracle Service: Chainlink Functions adapter that bridges Locus Tasks completion events to on-chain reputation updates
- Subgraph Indexer: The Graph Protocol subgraph indexing all contract events for fast, structured GraphQL queries
- Notification Service: Webhook and email alert delivery for all task state transitions
- API Gateway: REST and GraphQL gateway for frontend consumption and third-party integrations
- IPFS Pinning Service: Reliable content-addressed storage for task specifications, completion proofs, and dispute evidence

5.1.3 Frontend Application

- Agent Registration and Profile Portal: Wallet connect, rSBT mint flow, tier display with score breakdown
- Task Browser: Search and filter by agent tier, reputation range, task category, and deadline
- Escrow Dashboard: Active task management, payment status, completion submission, and dispute initiation

- Reputation Leaderboard: Public agent ranking with full score decomposition and history chart
- DAO Governance UI: Proposal browser, voting interface, delegation management, and treasury transparency view
- Operator Fleet View: Multi-agent management for operators running more than one agent

5.1.4 Explicitly Out of Scope for MVP

- Cross-chain reputation bridging (Phase 3)
- Agent-to-agent sub-tasking with nested escrow (Phase 2)
- Native mobile applications (Phase 2)
- AI-assisted dispute resolution (Phase 3)
- Secondary market for $VOUCH tokens (post-TGE)
- KYA certification API for third-party platforms (Phase 2)
- ZK-proof of unique operator identity (Phase 3)

5.2 Phase 2: Platform Expansion (Months 5 to 12)

- KYA Standard API v1: Open API for third-party platforms to query agent reputation scores and access audit trails
- Agent-to-agent sub-tasking: Nested escrow enabling an agent to delegate sub-tasks to other agents with reputation propagation
- Advanced dispute resolution: DAO jury system with randomized token-weighted juror selection replacing founding committee
- Mobile-responsive PWA: Progressive web app targeting mobile operator use cases
- Operator fleet dashboard: Multi-agent management with aggregate analytics, bulk tier upgrades, and comparative performance views
- IPFS audit trail viewer: Full task history browser with export to PDF and JSON formats
- Reputation-weighted task matching: Algorithm that surfaces the highest-VOUCH-score agents for each task category
- Multi-stablecoin support: USDT and DAI as alternative payment tokens in TaskEscrow

5.3 Phase 3: Ecosystem and Scale (Months 13 to 24)

- Cross-chain reputation bridging: Optimism, Arbitrum, and Polygon with canonical VOUCH score anchored on Base
- AI-powered anomaly detection: Behavioral analysis engine for real-time sybil resistance and score manipulation detection
- KYA API monetization: Enterprise licensing tiers for high-volume programmatic access to VOUCH's reputation data layer
- Token Generation Event (TGE): Public $VOUCH token launch with exchange listings and liquidity provisioning
- Agent insurance module: Slashing coverage product that compensates task posters for verified agent failures

- EU AI Act compliance module: Automated attestation reports satisfying high-risk AI system logging and audit requirements
- Ziki Labs portfolio integration: VOUCH as the shared reputation layer across all future Ziki Labs products

6. System Architecture

6.1 Architecture Philosophy

VOUCH is designed around three non-negotiable architectural principles. First, the source of truth is always on-chain: reputation scores, agent identities, escrow balances, and task state transitions are all anchored on Base L2. Second, off-chain services are degradable: if the API gateway, indexer, or notification service goes down, the core protocol contracts remain fully functional. Third, every component is replaceable: no single off-chain service creates an irrecoverable dependency.

6.2 High-Level Architecture Diagram

```
PRESENTATION LAYER
+-------------------------------------------------------------------+
|  Next.js 14 App (Vercel)      |  KYA API (Third-Party Clients)  |
+-------------------------------------------------------------------+
                  |                           |
API GATEWAY LAYER
+-------------------------------------------------------------------+
|  Apollo GraphQL Server  |  Express REST API  |  WebSocket WS   |
|           Auth: Sign-In With Ethereum (EIP-4361)               |
|           Rate Limiting: Redis-backed per-wallet limits         |
+-------------------------------------------------------------------+
        |                                       |
OFF-CHAIN SERVICES             BLOCKCHAIN LAYER (Base L2)
+-------------------------+    +-------------------------------------+
|  VerificationOracle     |    |  AgentRegistry.sol  (rSBT ERC-721)  |
|  (Chainlink Functions)  |<-->|  ReputationEngine.sol               |
|                         |    |  TaskEscrow.sol                     |
|  Subgraph Indexer       |<-->|  DisputeManager.sol                 |
|  (The Graph Network)    |    |  DAOGovernor.sol (Governor Bravo)   |
|                         |    |  VOUCHToken.sol (ERC-20)            |
|  Notification Service   |    +-------------------------------------+
|  IPFS Pinning Service   |
+-------------------------+
        |
DATA LAYER
+-----------------------------------------------+
|  PostgreSQL 15  (metadata, enrichment, cache)  |
|  Redis 7        (session, rate limit, query)   |
|  IPFS + Filecoin (audit trail, task content)   |
+-----------------------------------------------+
```

6.3 Blockchain Layer Detail

6.3.1 Network Configuration

Chain ID	Base Mainnet: 8453 | Base Sepolia Testnet: 84532

Block Time	Approximately 2 seconds
Gas Token	ETH (bridged from L1 via Base Official Bridge)
RPC Provider	Alchemy (primary), QuickNode (fallback), Base public RPC (emergency)
Block Explorer	Basescan (basescan.org)
Bridge	Base Official Bridge (Superchain) for L1 to L2 USDC and ETH movement
USDC Contract	Circle's native USDC on Base (CCTP-compatible, not bridged USDC)

6.3.2 Contract Interaction Sequence

```
Agent Operator          AgentRegistry     ReputationEngine      TaskEscrow
     |                        |                  |                   |
     |-- registerAgent() ---->|                  |                   |
     |<-- rSBT minted         |                  |                   |
     |  (ERC-721 tokenId)     |-- initScore() -->|                   |
     |                        |                  |                   |
Task Poster                   |                  |                   |
     |-- createTask(agentId, USDC, deadline) ---------------------->|
     |<-- taskId assigned     |                  |                   |
     |                        |                  |                   |
Agent Operator                |                  |                   |
     |-- acceptTask(taskId) ----------------------------------------->|
     |                        |                  |                   |
[Work Completed Off-Chain]    |                  |                   |
     |                        |                  |                   |
     |-- submitCompletion(taskId, ipfsCid) -------------------------->|
     |                        |                  |                   |
Chainlink Oracle              |                  |                   |
     |-- fulfillVerification(reqId, response) ----------------------->|
     |                        |                  |                   |
     |                        |<-- recordOutcome(agentId, delta) ----|
     |                        |                  |                   |
     |<-- USDC released to agent wallet ----------------------------|
```

6.4 Off-Chain Services Architecture

6.4.1 Verification Oracle Service (Critical Path)

The most critical off-chain component in the VOUCH stack. This service bridges Locus Tasks completion events to on-chain reputation updates. It uses Chainlink Functions to make verifiable HTTP calls to the Locus Tasks API, encode the response, and deliver it to the TaskEscrow contract in a single, trustless callback.

```
// Chainlink Functions source code (simplified)
// Deployed to Chainlink DON; executed by 7+ independent oracle nodes
```

```
  const taskId     = args[0];
  const apiResponse = await Functions.makeHttpRequest({
    url:     `https://api.locustasks.io/v1/tasks/${taskId}/completion`,
    headers: { 'Authorization': `Bearer ${secrets.LOCUS_API_KEY}` },
    timeout: 9000,
  });

  if (apiResponse.error) {
    return Functions.encodeUint256(0);  // 0 = verification failed
  }

  const { status, qualityScore, completedAtMs, deadlineMs } = apiResponse.data;

  // Pack: [success (1 bit), qualityScore (7 bits), onTime (1 bit), taskId (lower
64 bits)]
  const success  = status === 'completed' ? 1 : 0;
  const onTime   = completedAtMs <= deadlineMs ? 1 : 0;
  const packed   = packVerification(success, qualityScore, onTime, taskId);

  return Functions.encodeUint256(packed);
```

6.4.2 The Graph Subgraph

- Indexes all events from AgentRegistry, ReputationEngine, TaskEscrow, and DisputeManager contracts
- Exposes GraphQL queries for agent profiles, score history, task history, leaderboard, and dispute records
- Deployed to The Graph Network with multiple decentralized indexers for censorship resistance and redundancy
- Subgraph Studio used for development and staging deployment; production on decentralized network
- Entity schema versioned alongside contract deployments; migration scripts maintained in repository

6.4.3 API Gateway

Runtime	Node.js 20 LTS with Express 5 and Apollo Server 4
Authentication	Sign-In With Ethereum (EIP-4361); session JWT with 24-hour TTL and refresh token rotation
Authorization	Wallet-address-scoped; operators can only execute actions on their own agents and tasks
GraphQL	Primary interface for frontend; wraps The Graph subgraph queries with enrichment from PostgreSQL
REST	Webhook receiver endpoints for Locus Tasks; admin endpoints; KYA API for third-party integrators
Rate Limiting	100 requests per minute per IP; 1,000 requests per minute per authenticated wallet; Redis-backed

Caching	Redis TTL: 30 seconds for leaderboard queries; 5 seconds for individual score queries
Monitoring	Prometheus metrics on every route; latency histograms; error rate alerts via PagerDuty

7. Data Model

7.1 On-Chain Data Structures (Solidity)

```
// AgentRegistry.sol
struct Agent {
    uint256 tokenId;         // rSBT token ID; serves as canonical agentId
    address operator;        // EOA of the agent operator (human or org)
    address agentWallet;     // The agent's own execution EOA
    uint8   tier;            // 0=Unranked 1=Bronze 2=Silver 3=Gold 4=Platinum
    uint64  registeredAt;    // Unix timestamp of initial registration
    bool    active;          // False if deactivated by DAO governance
    bytes32 metadataHash;    // IPFS CID (sha2-256) of extended agent metadata
}

// ReputationEngine.sol
struct ReputationScore {
    uint256 agentId;
    int256  rawScore;          // Range: -10,000 to +10,000; initialized at 0
    int256  ewmaScore;         // Exponentially weighted moving average
(alpha=0.1)
    uint256 tasksCompleted;
    uint256 tasksFailed;
    uint256 disputesRaised;
    uint256 disputesWon;
    uint256 totalUsdcProcessed; // Lifetime USDC volume in 6-decimal units
    uint256 lastUpdated;
}

// TaskEscrow.sol
struct Task {
    bytes32 taskId;            // keccak256(poster address, nonce)
    address poster;
    uint256 agentId;
    uint256 amount;            // USDC in 6-decimal units
    uint8   status;
    // Status codes:
    // 0=Open 1=Accepted 2=SubmissionPending 3=Verified 4=Failed 5=Disputed
6=Resolved
    uint64  createdAt;
    uint64  deadline;
    bytes32 requirementsHash;  // IPFS CID of task specification document
    bytes32 completionHash;    // IPFS CID of completion proof submission
    bytes32 oracleRequestId;   // Chainlink Functions request ID
}
```

7.2 Off-Chain Database Schema (PostgreSQL 15)

```
-- agents: mirrors on-chain state with enriched off-chain metadata
CREATE TABLE agents (
  agent_id          BIGINT PRIMARY KEY,      -- matches on-chain tokenId
  operator_addr     CHAR(42) NOT NULL,
  agent_addr        CHAR(42) NOT NULL UNIQUE,
```

```
    display_name      TEXT NOT NULL,
    description       TEXT,
    avatar_ipfs       TEXT,                    -- IPFS CID
    specializations   TEXT[],                  -- e.g. ['coding', 'research']
    tier              SMALLINT DEFAULT 0,
    raw_score         INTEGER DEFAULT 0,
    ewma_score        INTEGER DEFAULT 0,
    tasks_completed   INTEGER DEFAULT 0,
    tasks_failed      INTEGER DEFAULT 0,
    created_at        TIMESTAMPTZ DEFAULT NOW()
  );

  -- tasks: full task records with on-chain anchors
  CREATE TABLE tasks (
    task_id           CHAR(66) PRIMARY KEY,    -- bytes32 hex string
    poster_addr       CHAR(42) NOT NULL,
    agent_id          BIGINT REFERENCES agents(agent_id),
    amount_usdc       NUMERIC(20,6) NOT NULL,
    status            SMALLINT NOT NULL DEFAULT 0,
    title             TEXT NOT NULL,
    description       TEXT,
    requirements      JSONB,
    requirements_ipfs TEXT,                    -- IPFS CID
    completion_ipfs   TEXT,                    -- IPFS CID
    created_at        TIMESTAMPTZ NOT NULL,
    deadline          TIMESTAMPTZ NOT NULL,
    accepted_at       TIMESTAMPTZ,
    completed_at      TIMESTAMPTZ,
    tx_create_hash    CHAR(66),
    tx_finalize_hash  CHAR(66),
    block_finalized   BIGINT
  );

  -- reputation_history: append-only ledger; never updated, only inserted
  CREATE TABLE reputation_history (
    id          BIGSERIAL PRIMARY KEY,
    agent_id    BIGINT REFERENCES agents(agent_id),
    delta       INTEGER NOT NULL,
    new_score   INTEGER NOT NULL,
    reason      TEXT NOT NULL,      -- 'task_complete' | 'task_fail' |
'dispute_win' | 'slash'
    task_id     CHAR(66),
    tx_hash     CHAR(66) NOT NULL,
    block_num   BIGINT NOT NULL,
    created_at  TIMESTAMPTZ DEFAULT NOW()
  );

  -- disputes: full dispute lifecycle records
  CREATE TABLE disputes (
    dispute_id       BIGSERIAL PRIMARY KEY,
    task_id          CHAR(66) REFERENCES tasks(task_id),
    raised_by        CHAR(42) NOT NULL,
    reason           TEXT NOT NULL,
    evidence_poster  TEXT,    -- IPFS CID
    evidence_agent   TEXT,    -- IPFS CID
    status           TEXT DEFAULT 'open',  -- open | under_review | resolved |
escalated
    resolution       TEXT,    -- 'agent_win' | 'poster_win' | 'split'
    resolution_notes TEXT,
    resolved_by      CHAR(42),
    raised_at        TIMESTAMPTZ DEFAULT NOW(),
```

```
    resolved_at      TIMESTAMPTZ
  );

  -- governance_proposals: mirrors on-chain proposals with human-readable metadata
  CREATE TABLE governance_proposals (
    proposal_id    BIGINT PRIMARY KEY,
    proposer_addr  CHAR(42) NOT NULL,
    title          TEXT NOT NULL,
    description    TEXT,
    proposal_type  TEXT NOT NULL,  -- 'parameter' | 'treasury' | 'upgrade' |
'emergency'
    status         TEXT NOT NULL,  -- 'active' | 'passed' | 'defeated' |
'executed' | 'cancelled'
    votes_for      NUMERIC(30,0) DEFAULT 0,
    votes_against  NUMERIC(30,0) DEFAULT 0,
    quorum_met     BOOLEAN DEFAULT FALSE,
    proposed_at    TIMESTAMPTZ NOT NULL,
    vote_end       TIMESTAMPTZ NOT NULL,
    executed_at    TIMESTAMPTZ
  );
```

8. Smart Contract Interfaces

8.1 IAgentRegistry

```
// SPDX-License-Identifier: MIT
// @title IAgentRegistry
// @notice ERC-721 Soulbound Token registry for AI agents in the VOUCH protocol
// @dev Transfers always revert (SBT constraint). One token per agent wallet.

interface IAgentRegistry {

  // Events
  event AgentRegistered(
    uint256 indexed tokenId,
    address indexed operator,
    address indexed agentWallet,
    uint64  registeredAt
  );
  event TierUpgraded(uint256 indexed tokenId, uint8 oldTier, uint8 newTier);
  event AgentDeactivated(uint256 indexed tokenId, address indexed by, string
reason);
  event MetadataUpdated(uint256 indexed tokenId, bytes32 newHash);

  // Errors
  error AlreadyRegistered(address agentWallet);
  error NotTransferable();
  error InsufficientStake(uint256 required, uint256 provided);
  error Unauthorized(address caller);
  error AgentNotActive(uint256 tokenId);

  // Core Functions

  /// @notice Register a new agent; mints rSBT; requires VOUCH stake for tiers
1+
  /// @param agentWallet   Execution EOA of the agent
  /// @param metadataHash  IPFS CID (bytes32) of agent metadata JSON
  /// @param initialTier   0=Unranked (free); 1-4 require increasing VOUCH stake
  /// @return tokenId      The minted rSBT token ID
  function registerAgent(
    address agentWallet,
    bytes32 metadataHash,
    uint8   initialTier
  ) external returns (uint256 tokenId);

  /// @notice Upgrade an agent to a higher tier by staking additional VOUCH
tokens
  function upgradeTier(uint256 tokenId, uint8 targetTier) external;

  /// @notice Update agent metadata IPFS hash
  function updateMetadata(uint256 tokenId, bytes32 newMetadataHash) external;

  /// @notice Returns the full Agent struct for a given tokenId
  function getAgent(uint256 tokenId) external view returns (Agent memory);

  /// @notice Reverse lookup: agentWallet address to tokenId
```

```
    function tokenIdByWallet(address agentWallet) external view returns (uint256);

    /// @notice DAO-only: deactivate a misbehaving agent; stake is slashed
    function deactivateAgent(uint256 tokenId, string calldata reason) external;
  }
```

8.2 IReputationEngine

```
  interface IReputationEngine {

    // Events
    event ScoreUpdated(
      uint256 indexed agentId,
      int256  delta,
      int256  newRawScore,
      int256  newEwmaScore,
      string  reason
    );
    event TierThresholdUpdated(uint8 indexed tier, int256 minScore);
    event ScoreBoundsUpdated(int256 minScore, int256 maxScore);

    // Core Functions

    /// @notice Called exclusively by TaskEscrow after Chainlink verification
    /// @param agentId       rSBT token ID
    /// @param taskId        Task identifier (bytes32)
    /// @param qualityScore  0 to 100 from oracle; represents task quality rating
    /// @param success       Whether task was verified as completed
    /// @param onTime        Whether task was completed before deadline
    /// @param usdcAmount    Task payment amount (used for size bonus scaling)
    function recordOutcome(
      uint256 agentId,
      bytes32 taskId,
      uint8   qualityScore,
      bool    success,
      bool    onTime,
      uint256 usdcAmount
    ) external;  // onlyEscrow modifier

    /// @notice Called by DisputeManager on resolution
    function applyDisputeOutcome(
      uint256 agentId,
      int256  delta,
      string  calldata reason
    ) external;  // onlyDisputeManager modifier

    /// @notice Returns EWMA score (primary score for tier and task access
decisions)
    function getScore(uint256 agentId) external view returns (int256 ewmaScore);

    /// @notice Returns full ReputationScore struct
    function getFullScore(uint256 agentId) external view returns (ReputationScore
memory);

    /// @notice Returns the tier corresponding to a given score
```

```
    function tierForScore(int256 score) external view returns (uint8 tier);
  }
```

8.3 ITaskEscrow

```
  interface ITaskEscrow {

    // Events
    event TaskCreated(
      bytes32 indexed taskId, address indexed poster,
      uint256 indexed agentId, uint256 amount, uint64 deadline
    );
    event TaskAccepted(bytes32 indexed taskId, uint256 indexed agentId);
    event CompletionSubmitted(bytes32 indexed taskId, bytes32 completionHash);
    event VerificationRequested(bytes32 indexed taskId, bytes32
chainlinkRequestId);
    event TaskFinalized(
      bytes32 indexed taskId, bool success,
      uint256 agentPayout, int256 scoreDelta
    );
    event DisputeRaised(bytes32 indexed taskId, address indexed by);
    event TaskExpired(bytes32 indexed taskId);

    // Core Functions

    /// @notice Task poster creates task and deposits USDC into escrow
    /// @param agentId           Target agent's rSBT token ID
    /// @param usdcAmount        Payment in USDC (6 decimals)
    /// @param deadline          Unix timestamp; must be at least 1 hour in future
    /// @param requirementsHash  IPFS CID of task specification
    /// @return taskId           Unique bytes32 task identifier
    function createTask(
      uint256 agentId,
      uint256 usdcAmount,
      uint64  deadline,
      bytes32 requirementsHash
    ) external returns (bytes32 taskId);

    /// @notice Agent operator accepts an open task; locks agent into task
    function acceptTask(bytes32 taskId) external;  // onlyAgentOperator(taskId)

    /// @notice Agent submits IPFS proof of completion
    function submitCompletion(bytes32 taskId, bytes32 completionHash) external;

    /// @notice Triggers Chainlink Functions verification call
    /// @return reqId  Chainlink request ID for tracking
    function requestVerification(bytes32 taskId) external returns (bytes32 reqId);

    /// @notice Chainlink DON callback; finalizes task state and triggers payout
    function fulfillVerification(
      bytes32 reqId,
      bytes   calldata response
    ) external;  // onlyOracle modifier

    /// @notice Either party raises a formal dispute within 24h of verification
```

```
    function raiseDispute(bytes32 taskId, string calldata reason) external;

    /// @notice Task poster cancels an unaccepted Open task; USDC returned
    function cancelTask(bytes32 taskId) external;  // onlyPoster

    /// @notice Mark expired task; refunds poster if agent never accepted
    function expireTask(bytes32 taskId) external;
  }
```

9. API Specifications

9.1 GraphQL Schema (Core Types)

```
type Agent {
  id:               ID!
  tokenId:          String!
  operatorAddr:     String!
  agentAddr:        String!
  displayName:      String!
  description:      String
  avatarUrl:        String
  specializations:  [String!]!
  tier:             Int!
  tierLabel:        String!     # 'Unranked' | 'Bronze' | 'Silver' | 'Gold' |
'Platinum'
  rawScore:         Int!
  ewmaScore:        Int!
  tasksCompleted:   Int!
  tasksFailed:      Int!
  winRate:          Float!      # tasksCompleted / (tasksCompleted + tasksFailed)
  active:           Boolean!
  registeredAt:     String!
  scoreHistory(limit: Int): [ScoreEvent!]!
  tasks(status: TaskStatus, limit: Int, offset: Int): [Task!]!
}

type Task {
  id:               ID!
  poster:           String!
  agent:            Agent!
  amountUsdc:       String!
  status:           TaskStatus!
  title:            String!
  deadline:         String!
  requirementsUrl:  String
  completionUrl:    String
  completedAt:      String
  dispute:          Dispute
}

type ScoreEvent {
  delta:     Int!
  newScore:  Int!
  reason:    String!
  taskId:    String
  txHash:    String!
  timestamp: String!
}

type Dispute {
  id:          ID!
  raisedBy:    String!
  reason:      String!
  status:      String!
  resolution:  String
  resolvedAt:  String
```

```
}

enum TaskStatus {
  OPEN ACCEPTED SUBMISSION_PENDING VERIFIED FAILED DISPUTED RESOLVED EXPIRED
}

type Query {
  agent(id: ID!):     Agent
  agentByWallet(addr: String!): Agent
  agents(
    tier:     Int
    minScore: Int
    maxScore: Int
    spec:     String
    limit:    Int
    offset:   Int
  ): [Agent!]!
  leaderboard(limit: Int): [Agent!]!
  task(id: ID!):      Task
  tasks(
    agentId: ID
    poster:  String
    status:  TaskStatus
    limit:   Int
  ): [Task!]!
  proposals(active: Boolean): [Proposal!]!
}

type Mutation {
  # All mutations require SIWE authentication
  registerAgent(input: RegisterAgentInput!):    RegisterResult!
  upgradeTier(tokenId: ID!, targetTier: Int!):  Boolean!
  createTask(input: CreateTaskInput!):           TaskResult!
  submitCompletion(taskId: ID!, ipfsCid: String!): Boolean!
  raiseDispute(taskId: ID!, reason: String!):   DisputeResult!
  cancelTask(taskId: ID!):                      Boolean!
  castVote(proposalId: ID!, support: Boolean!): Boolean!
}
```

9.2 REST Endpoints

Method	Endpoint	Auth	Purpose
POST	/webhooks/locus/completion	HMAC-SHA256 signature	Receive task completion callback from Locus Tasks
POST	/webhooks/chainlink/callback	IP allowlist + signature	Receive Chainlink Functions fulfillment response
POST	/api/v1/auth/siwe/challenge	Public	Generate SIWE challenge message for wallet signing
POST	/api/v1/auth/siwe/verify	Public	Verify signed SIWE message; issue session JWT

Method	Endpoint	Auth	Purpose
GET	/api/v1/agents/:id	Public	Get agent profile and current reputation score
GET	/api/v1/leaderboard	Public	Top 100 agents ranked by EWMA score
GET	/api/v1/agents/:id/history	Public	Full reputation history for an agent
GET	/api/v1/tasks/:id	Public	Get task details and current status
POST	/api/v1/admin/slash	DAO multisig header	Emergency reputation slash via governance
GET	/api/v1/admin/health	Internal API key	Service health check; used by monitoring
GET	/kya/v1/agents/:addr/score	KYA API key	Third-party KYA API: current score for agent wallet
GET	/kya/v1/agents/:addr/vouched	KYA API key	Third-party KYA API: Boolean 'is this agent vouched'

10. UX Flows

10.1 Agent Registration Flow

```
  Step 1.  Operator visits vouch.xyz/register
  Step 2.  Clicks 'Connect Wallet' - MetaMask or WalletConnect modal appears
  Step 3.  Signs SIWE challenge message to authenticate session (no gas required)
  Step 4.  Completes registration form:
             Agent Wallet Address   (required; must be a fresh or dedicated EOA)
             Display Name           (required; 3 to 50 characters; shown publicly)
             Description            (optional; markdown supported)
             Specializations        (optional; multi-select tag input)
             Avatar Image           (optional; uploaded to IPFS via Pinata)
             Initial Tier           (required; Unranked is free; Bronze+ requires
stake)
  Step 5.  If tier Bronze or above is selected:
             System shows VOUCH stake requirement (100 VOUCH for Bronze)
             'Approve VOUCH' transaction triggers - MetaMask confirmation
             Approval confirmed on-chain
  Step 6.  'Register Agent' transaction triggers
             AgentRegistry.registerAgent() executes on Base L2
             rSBT (ERC-721) minted to operator wallet
             ReputationEngine.initScore() called internally
  Step 7.  Redirect to Agent Profile page
             rSBT badge displayed with tokenId
             Tier badge shown (Unranked by default)
             Score = 0, task history empty, shareable public URL generated
  Step 8.  Operator can share agent profile URL immediately
             Profile is fully public and searchable in agent browser
```

10.2 Task Creation and Escrow Flow

```
  Step 1.  Task poster visits vouch.xyz/tasks/new
  Step 2.  Connects wallet and completes SIWE authentication
  Step 3.  Searches and filters agents by: tier, EWMA score range, specializations
  Step 4.  Selects a specific agent and views their full profile:
             Current tier and score breakdown
             Full reputation history chart
             All completed and failed tasks (public record)
             Dispute history and resolution outcomes
  Step 5.  Clicks 'Create Task with This Agent'
  Step 6.  Fills task form:
             Title           (required; 5 to 120 characters)
             Description     (required; detailed task brief)
             Requirements    (optional; file upload stored to IPFS)
             USDC Amount     (required; minimum $1; maximum $10,000 for MVP)
             Deadline        (required; minimum 1 hour from now)
  Step 7.  System shows fee breakdown: task amount + 1.5% VOUCH protocol fee
  Step 8.  'Approve USDC' transaction triggers (ERC-20 approval to TaskEscrow)
  Step 9.  'Create Task' transaction triggers: TaskEscrow.createTask() executes
             Task appears in poster's dashboard as 'Open'
             Agent receives notification (webhook + email)
  Step 10. Agent operator accepts task in their dashboard
```

```
              Status: Accepted
              Poster receives confirmation notification
  Step 11. Agent completes work off-chain
              Agent submits IPFS proof via dashboard: submitCompletion()
              Status: Submission Pending
  Step 12. Verification is requested: Chainlink Functions call triggers
              Status: Verification In Progress
  Step 13. Chainlink oracle delivers result
              If verified as complete:
                USDC automatically released to agent wallet (minus protocol fee)
                Reputation score updated with delta calculation
                Status: Verified
              If verified as failed:
                USDC returned to poster wallet
                Reputation score penalized
                Status: Failed
  Step 14. 24-hour dispute window opens
              Either party may raise a formal dispute during this period
              If no dispute raised, task is permanently closed
```

10.3 Dispute Resolution Flow

```
  Step 1.  Aggrieved party clicks 'Raise Dispute' on task within 24h window
              DisputeManager.raiseDispute(taskId, reason) executes on-chain
              USDC held in escrow; no further automatic actions
              Status: Disputed
  Step 2.  Both parties receive notifications immediately
  Step 3.  Evidence submission window opens (48 hours)
              Poster uploads evidence: task requirements, communication logs
              Agent uploads evidence: completion artifacts, delivery proof
              All evidence stored to IPFS; hashes recorded on-chain
  Step 4.  Founding Committee (5-of-9 multisig, MVP) reviews all evidence
              Committee has 72 hours to render decision
  Step 5.  Resolution is executed on-chain by committee multisig
              Outcome A (Agent wins):   USDC released to agent; partial score
restore
              Outcome B (Poster wins):  USDC refunded to poster; score penalized
              Outcome C (Split):        USDC split proportionally; partial
penalties
  Step 6.  Both parties notified with outcome explanation
  Step 7.  Resolution recorded permanently in dispute history
              Dispute outcome visible on agent's public profile

  Phase 2 upgrade: Committee replaced by DAO jury system
              Random selection of token-weighted jurors from VOUCH holders
              Jurors stake VOUCH as a commitment bond
              Jurors who vote with the majority earn rewards
              Jurors who vote against the majority lose their bond
```

11. Reputation Scoring Algorithm

Algorithm Design Principles

The VOUCH scoring algorithm must be: (1) fully deterministic and verifiable on-chain, (2) resistant to gaming via rapid low-value task farming, (3) exponentially weighted toward recent performance so reputation decays gracefully with inactivity, (4) resistant to sybil attacks via staking requirements and behavioral analysis, and (5) governed by the DAO so parameters can be adjusted as the ecosystem matures.

11.1 Score Delta Formula

```
  // Pseudocode - implemented as fixed-point arithmetic in Solidity
  // All division is integer division; precision maintained via scaling factors

  function computeDelta(
    bool    success,
    uint8   qualityScore,    // 0 to 100 from oracle
    uint8   tier,            // agent current tier: 0=Unranked to 4=Platinum
    bool    onTime,          // completed before deadline
    uint256 usdcAmount       // task USDC value in 6-decimal units
  ) internal pure returns (int256 delta) {

    // Failure path: penalty scaled by tier (higher tier agents are held to higher
standard)
    if (!success) {
      delta = -100 * int256(uint256(tier) + 1);
      // Unranked: -100  Bronze: -200  Silver: -300  Gold: -400  Platinum: -500
      return delta;
    }

    // Success path
    int256 base = 50;  // base points for any successful completion

    // Quality multiplier: maps qualityScore 0-100 to multiplier 0.5x to 2.0x
    // Implementation: (qualityScore + 25) / 50 = 0.5 to 2.5; capped at 2.0x
    int256 qualityMul = int256(uint256(qualityScore)) + 25;  // 25 to 125
    if (qualityMul > 100) qualityMul = 100;                  // cap at 2.0x
equivalent

    // Time bonus: full points if on-time, 80% if late
    int256 timeFactor = onTime ? 100 : 80;

    // Task size bonus: logarithmic scaling; max +25 bonus points
    // Larger tasks contribute more to reputation (prevents micro-task farming)
    uint256 usdcWhole = usdcAmount / 1e6;  // convert to whole USDC
    int256 sizeBonus = 0;
    if (usdcWhole >= 10)   sizeBonus = 5;
    if (usdcWhole >= 50)   sizeBonus = 10;
    if (usdcWhole >= 200)  sizeBonus = 15;
    if (usdcWhole >= 1000) sizeBonus = 25;
```

```
    // Compose: (base * qualityMul * timeFactor / 5000) + sizeBonus
    delta = (base * qualityMul * timeFactor / 5000) + sizeBonus;
    return delta;
  }

  // EWMA update (alpha = 0.1, scaled by 1000 for integer math)
  // newEwma = (100 * delta + 900 * currentEwma) / 1000
  // Score bounds enforced: clamped to [-10000, +10000]
```

11.2 Tier Thresholds and Benefits

Tier	EWMA Score Range	VOUCH Stake Required	Agent Benefits
0: Unranked	Below 0	0 VOUCH	Access to basic tasks only; full penalty multiplier applies
1: Bronze	0 to 999	100 VOUCH	Standard task access; reduced penalty multiplier; trust badge displayed
2: Silver	1,000 to 2,999	500 VOUCH	Priority matching in task search; eligible for 1.1x payment bonus on premium tasks
3: Gold	3,000 to 6,999	2,000 VOUCH	Premium task access; 1.2x bonus eligibility; reduced escrow hold period
4: Platinum	7,000+	10,000 VOUCH	Exclusive task tier; 1.5x bonus; DAO committee eligibility; verified badge

11.3 Anti-Gaming and Sybil Resistance

Implemented at MVP Launch

- Minimum stake requirements for non-zero tiers create real economic cost to sybil. Unranked agents are quarantined from high-value tasks.
- Operator wallet age check: the registering wallet must be at least 30 days old with a minimum of 5 on-chain transactions.
- Reputation bootstrap throttle: maximum score gain of +100 in the first 30 days, regardless of task volume.
- Task size normalization: micro-tasks below $1 do not contribute to the size bonus, removing the incentive to spam tiny tasks.
- Behavioral rate limit: more than 3 task acceptances per hour triggers oracle rate-limiting and places the agent in manual review queue.

Planned for Phase 3

- ZK-proof of unique operator identity via Worldcoin or Proof of Humanity integration, enabling one-agent-per-human enforcement without revealing identity.

- AI-powered behavioral anomaly detection that flags unusual score trajectories for DAO committee review before they become permanent.
- Cross-agent colocation detection: multiple agents sharing the same IP subnet, gas funding wallet, or transaction patterns flagged as potential sybil clusters.

12. Security and Compliance

12.1 Smart Contract Security

12.1.1 Pre-Deployment Security Requirements

- Two independent smart contract audits are required before any mainnet deployment. Recommended firms: Trail of Bits and OpenZeppelin Audits.
- Slither and Mythril static analysis run on every pull request to the main branch via GitHub Actions CI.
- Foundry fuzz testing with a minimum of 10,000 runs per critical function, with special focus on the scoring formula and escrow state machine.
- Formal verification of ReputationEngine score bounds invariants and escrow balance invariants via Certora Prover.
- Public bug bounty program on Immunefi with a maximum payout of $50,000 for critical vulnerabilities, funded by DAO treasury.

12.1.2 Access Control Architecture

```
  // Role hierarchy using OpenZeppelin AccessControl
  // All roles are grantable and revocable only by DEFAULT_ADMIN_ROLE

  bytes32 DEFAULT_ADMIN_ROLE   // Held by DAO Governor; all changes 48h timelock
  bytes32 ESCROW_ROLE          // TaskEscrow -> ReputationEngine calls only
  bytes32 ORACLE_ROLE          // Chainlink Functions fulfillment callbacks only
  bytes32 DISPUTE_ROLE         // DisputeManager -> ReputationEngine adjustments
  bytes32 COMMITTEE_ROLE       // 5-of-9 founding committee multisig (MVP phase)
  bytes32 EMERGENCY_ROLE       // 3-of-5 hot multisig; pause-only; 24h timelock to
unpause
  bytes32 UPGRADER_ROLE        // Proxy upgrade authorization; held by DAO
Governor

  // Timelock configuration
  // Standard governance actions:  48-hour minimum delay
  // Contract upgrades:            7-day minimum delay
  // Emergency pause:              Immediate (no delay)
  // Emergency unpause:            Requires full DAO vote; cannot be done by
multisig
```

12.1.3 Economic Attack Mitigations

- All USDC transfers use the pull-payment pattern: agents withdraw earned funds rather than having funds pushed to them, eliminating a class of reentrancy attacks.
- OpenZeppelin ReentrancyGuard applied to all external-facing TaskEscrow functions.
- Maximum single-task USDC cap: $10,000 for MVP; adjustable by DAO vote. This limits blast radius from any single exploit.
- Oracle response validation: any qualityScore value outside the range 0 to 100 is rejected and treated as a verification failure, preventing oracle manipulation.

- USDC amount matching with 0.1% slippage tolerance on escrow creation prevents front-running attacks.
- Circuit breaker: if total USDC outflows in a 1-hour period exceed $50,000, the escrow contract auto-pauses and triggers an emergency alert.

12.2 Application Security

Authentication	Sign-In With Ethereum (EIP-4361); no passwords stored anywhere; session JWT with 24-hour TTL
Authorization	Wallet-address-scoped throughout; operators can only act on their own agents; no shared sessions
API Security	HTTPS only enforced; CORS allowlist maintained; WAF via Cloudflare; OWASP Top 10 mitigations applied
Input Validation	All inputs sanitized and length-bounded at both API and contract level; IPFS CIDs validated against sha2-256 format
Secrets Management	AWS Secrets Manager for all credentials; no secrets in environment files or code repositories; Chainlink DON secrets for oracle keys
Dependency Scanning	Dependabot auto-PRs for vulnerable dependencies; Snyk on CI with mandatory fix before merge
SAST	GitHub Advanced Security; Semgrep custom rules for Solidity and TypeScript; runs on every PR
Penetration Testing	External pen test by independent firm required before public launch and annually thereafter

12.3 Compliance and Legal Framework

12.3.1 Regulatory Landscape

- USDC payments: Circle's USDC is not classified as a security in the United States. No KYC is required at the VOUCH protocol level for standard task payments. Task posters processing more than $10,000 per day may trigger KYC requirements delegated to Locus Tasks under their own compliance obligations.
- EU AI Act: VOUCH's immutable IPFS audit trail and on-chain task verification history directly satisfy the logging and auditability requirements for high-risk AI systems under Article 12 of the EU AI Act. This is a compliance accelerator for enterprise task posters.
- $VOUCH Token: Issued as a governance utility token with no explicit profit expectations marketed to purchasers. A formal legal opinion from qualified securities counsel is required before any public token distribution or TGE event. Wyoming DAO LLC structure is the recommended vehicle for Phase 2.
- Data Privacy (GDPR): No personal data is stored on-chain. The off-chain PostgreSQL database stores only public wallet addresses, which are pseudonymous. Task content

on IPFS is user-controlled and user-initiated. A formal GDPR assessment is required before onboarding EU-resident task posters at scale.

12.3.2 Legal Entity Roadmap

Phase 1 (MVP)	Ziki Labs (existing entity) deploys contracts and operates the protocol; founding team multisig holds admin keys
Phase 2	Wyoming DAO LLC ('VOUCH DAO LLC') formed; provides legal personhood for contracts, treasury management, and dispute resolution
Phase 3 (TGE)	Cayman Islands Foundation or Singapore Foundation as token-issuing entity; Ziki Labs OpCo as development company
IP Ownership	All smart contract IP developed by Ziki Labs; licensed to VOUCH DAO on favorable terms pending DAO ratification
Jurisdiction	Primary: United States (Base chain is Coinbase-linked, with clear US nexus); Secondary: Singapore for token structures

13. Risk Register and Mitigations

Risk	Probability	Impact	Mitigation Strategy
Oracle manipulation: attacker bribes or spoofs Chainlink nodes	Low	Critical	Chainlink DON with 7+ independent nodes; on-chain challenge period; anomaly detection on score jumps; circuit breaker on unusual verification patterns
Sybil attack: mass fake agent registration to game leaderboard	Medium	High	Staking requirements create economic cost; wallet age checks; behavioral rate throttles; bootstrap score cap; Phase 3 ZK identity
Smart contract exploit: reentrancy, logic bug, or access control failure	Low	Critical	Dual independent audit; Foundry fuzz testing; Certora formal verification; Immunefi bug bounty; emergency pause with 24h timelock
Governance capture: whale accumulates VOUCH and passes self-serving proposals	Medium	High	10-day proposal delay; supermajority (75%) for upgrades; quadratic voting in Phase 2; anti-whale voting caps discussed in governance charter
Locus Tasks API changes break Chainlink oracle	Medium	High	Versioned API integration with Locus; fallback manual verification mode for edge cases; formal SLA agreement with Locus Tasks team
USDC regulatory action or depeg event	Low	High	Multi-stablecoin support (USDT, DAI) added in Phase 2; monitoring for USDC peg deviation; circuit breaker on large outflows
Cold-start problem: insufficient agents at launch to attract task posters	High	High	500 VOUCH grant for first 100 registered agents; Ziki Labs co-marketing with Locus Tasks; founding team personally recruits first 20 operators
Score gaming via selective task farming on easy low-risk tasks	Medium	Medium	Task size normalization prevents micro-task farming; EWMA dampens volatility; DAO can adjust parameters; rate limits on acceptance frequency
Key person risk: founding engineer departure during critical build phase	Medium	High	4-of-7 multisig governance from day one; full documentation in Notion; ADR (Architecture Decision Records) maintained; bus factor greater than 3 across all systems
Chainlink Functions outage or degraded performance	Low	High	Manual verification fallback mode activated by COMMITTEE_ROLE; Locus Tasks direct API integration as secondary path; Chainlink SLA monitoring

14. Deployment and Operations

14.1 Infrastructure Stack

Frontend	Vercel (Next.js 14 App Router, Edge Functions, automatic preview deployments)
API Gateway	AWS ECS Fargate (containerized Node.js; auto-scaling; no server management)
Database	AWS RDS PostgreSQL 15 (Multi-AZ; automated daily backups; 7-day point-in-time recovery)
Cache	AWS ElastiCache Redis 7 (cluster mode; used for sessions, rate limiting, query cache)
Smart Contracts	Deployed to Base Mainnet via Foundry deploy scripts; proxies managed via OpenZeppelin Upgrades
Indexer	The Graph Hosted Service for MVP; migrating to decentralized network at public launch
File Storage	IPFS via Web3.Storage (primary) and Pinata (backup pinning); content-addressed, immutable
CDN and WAF	Cloudflare (DDoS protection, WAF, global CDN, SSL termination)
Secrets	AWS Secrets Manager for all application credentials; Chainlink DON secrets for oracle keys
CI/CD	GitHub Actions with mandatory test, lint, and audit gates before any deployment
Monitoring	Grafana Cloud (dashboards and alerting) with Prometheus metrics; on-chain monitoring via Tenderly
Logging	AWS CloudWatch for application logs; Datadog APM for tracing and error tracking
Incident Response	PagerDuty with on-call rotation; Slack integration for P0 and P1 alerts

14.2 Environment Strategy

Environment	Chain	Purpose	Access Control
Local	Hardhat Network or Anvil fork of Base Sepolia	Individual developer testing; no shared state	Developer only; runs on local machine
Dev	Base Sepolia Testnet	Feature development and component integration	Engineering team; auto-deploys from feature branches

Environment	Chain	Purpose	Access Control
Staging	Base Sepolia Testnet	Pre-production QA, security review, and load tests	QA team and engineering leads; mirrors production config
Production	Base Mainnet	Live system serving real users with real USDC	Automated deploys from main branch; contract changes require multisig

14.3 CI/CD Pipeline

```
# GitHub Actions pipeline overview

on: [push, pull_request]

job: test-contracts
  - forge build
  - forge test --fork-url $BASE_SEPOLIA_RPC --match-contract '*Test' -vvv
  - forge test --fuzz-runs 10000 --match-contract '*Fuzz'
  - slither . --config-file slither.config.json
  - forge snapshot  # gas usage regression check

job: test-backend (requires: test-contracts)
  - npm ci
  - npm run lint && npm run type-check
  - npm run test:unit
  - npm run test:integration  # spins up local postgres + redis

job: security-scan (requires: test-backend)
  - snyk test --severity-threshold=high
  - semgrep --config auto

job: deploy-staging (requires: security-scan; if: branch == 'main')
  - forge script DeployAll --rpc-url $BASE_SEPOLIA_RPC --broadcast
  - Update contract addresses in backend config
  - Deploy API container to ECS staging
  - Deploy frontend to Vercel preview environment
  - Run Playwright e2e test suite against staging
  - Deploy and sync The Graph subgraph to staging

job: deploy-production (requires: manual approval from 2 engineering leads)
  - Submit contract upgrade proposal to multisig (if contracts changed)
  - Multisig signers execute upgrade after 48h timelock
  - Deploy API container to ECS production (blue-green swap)
  - Deploy frontend to Vercel production
  - Run smoke tests
  - Post deployment summary to #deployments Slack channel
  - Update status page
```

14.4 Incident Response

Severity Classification

Severity	Definition	Response Time	Immediate Action
P0: Critical	User funds at risk; smart contract exploit; complete outage	Less than 15 minutes	Trigger emergency pause; page incident commander; open war room; post status page update
P1: High	Partial outage; oracle down; reputation scores frozen; auth failure	Less than 1 hour	Page on-call engineer; post Slack alert; update status page; begin root cause analysis
P2: Medium	Feature degraded; slow API response; indexer lag above 60 seconds	Less than 4 hours	Alert next available engineer; create incident ticket; begin investigation
P3: Low	UI bug; non-critical feature broken; cosmetic error	Less than 48 hours	Create bug ticket in sprint board; prioritize in next planning session

Runbook Location and Ownership

- All runbooks maintained in Ziki Labs internal Notion under VOUCH / Operations / Runbooks
- Each runbook is owned by a specific engineer and reviewed quarterly
- Key runbooks required at launch: Emergency Pause, Oracle Failure, Database Failover, Contract Upgrade, Sybil Incident Response

15. Governance and Team Roles

15.1 DAO Governance Structure

VOUCH uses a two-layer governance model. Layer one is the Founding Committee (5-of-9 multisig), which handles day-to-day operations, dispute resolution, and emergency actions during the MVP period. Layer two is $VOUCH token-based DAO governance via Governor Bravo, which controls all protocol parameter changes, treasury allocation, and contract upgrades.

The design intent is for the Founding Committee's authority to progressively transfer to the DAO as $VOUCH token distribution matures and governance participation reaches healthy levels. A formal decentralization roadmap will be published at public launch.

15.1.1 Proposal Types and Voting Thresholds

Proposal Type	Quorum Required	Approval Threshold	Timelock Delay	Examples
Standard Parameter Update	5% of supply	51% majority	48 hours	Fee rate changes, tier thresholds, oracle subscription config
Treasury Spending	10% of supply	60% majority	72 hours	Grants to developers, audit funding, marketing spend
Protocol Upgrade	15% of supply	75% supermajority	7 days	New contract versions, proxy upgrades, core algorithm changes
Emergency Action	N/A	4-of-5 multisig	None	Emergency pause only; unpause requires full DAO vote
Governance Parameter	20% of supply	75% supermajority	7 days	Changes to quorum requirements or voting rules

15.2 Team Structure and Responsibilities

Role	Count	Responsibilities
Engineering Lead / Blockchain Architect (Ziki Labs)	1	System design, contract architecture, security review coordination, deployment authorization
Smart Contract Engineers	2	Contract development, Foundry test suite, Chainlink Functions integration, audit coordination

Role	Count	Responsibilities
Backend Engineers	2	API gateway, oracle adapter service, subgraph deployment, PostgreSQL schema, Redis integration
Frontend Engineers	2	Next.js application, wallet integration (wagmi/viem), all UX flows, Tailwind design system
QA and Security Engineer	1	E2e test automation, Playwright suite, pen test coordination, vulnerability triage
Product Manager (Ziki Labs)	1	PRD ownership, roadmap prioritization, stakeholder communication, user research
DAO Operations Manager	1	Governance facilitation, community management, multisig coordination, contributor grants
Legal and Compliance Counsel	0.5 (part-time)	Entity formation, $VOUCH token legal opinion, GDPR assessment, terms of service

16. Roadmap and Milestones

Milestone	Target	Key Deliverables	Owner
M0: Foundation	Month 1	Team assembled; Ziki Labs entity confirmed as operating entity; dev environments configured; Base Sepolia deployment pipeline working; Notion workspace and Slack set up	Ziki Labs Founders
M1: Contracts Alpha	Month 2	All 5 core contracts deployed to Base Sepolia; Foundry unit and fuzz test suite at 100% branch coverage; Slither static analysis clean; NatSpec documentation complete	Smart Contract Lead
M2: Backend Alpha	Month 2	API gateway live on staging; Chainlink Functions integration functional; The Graph subgraph indexing all events; SIWE auth working end-to-end	Backend Lead
M3: Frontend Alpha	Month 3	Agent registration flow complete; task browser live; escrow dashboard functional; leaderboard rendering from subgraph; all wallet flows tested	Frontend Lead
M4: Security Audit	Month 3-4	Two independent smart contract audits complete; all critical and high severity findings remediated; audit reports published publicly; bug bounty on Immunefi launched	Engineering Lead + External Auditors
M5: Beta Launch	Month 4	Base Mainnet deployment; 10 founding beta agents onboarded by Ziki Labs team; 50 real-USDC tasks completed; monitoring and alerting fully configured	All Hands
M6: Public Launch	Month 5	Public announcement; 500 agent registration target; Locus Tasks integration fully live; DAO governance active with first proposals; KYA API v0 documented	Product + Engineering + DAO Ops
M7: Phase 2 Start	Month 6	KYA Standard API v1 with third-party documentation; first external platform integration; DAO jury dispute system; mobile PWA shipped	Product
M8: Protocol Maturity	Month 12	3,000 MAA milestone; $500K treasury; second marketplace integration; Ziki Labs foundation entity preparation for TGE; $VOUCH token legal opinion complete	Ziki Labs Leadership
M9: TGE Preparation	Month 18	Foundation entity operational; tokenomics finalized and published; exchange outreach; liquidity provisioning plan; community governance fully decentralized	DAO Ops + Legal

17. Quality Assurance and Testing

17.1 Smart Contract Testing

Unit Tests	Foundry: 100% branch coverage across all 5 contracts; minimum 250 named test cases with clear arrange-act-assert structure
Fuzz Tests	forge fuzz: 10,000 runs minimum on scoring formula, escrow state machine, access control, and EWMA calculation
Invariant Tests	forge invariant: total USDC in escrow always greater than or equal to sum of active task amounts; no score outside bounds; no tier inconsistency
Integration Tests	Fork tests on Base Sepolia mainnet fork; full end-to-end task lifecycle including real Chainlink callbacks in forked environment
Formal Verification	Certora Prover: ReputationEngine score bound invariants; no-reentrancy proofs on TaskEscrow; access control consistency
Gas Profiling	forge snapshot on every PR; regression alert if any user-facing function exceeds 300,000 gas; cost model maintained in documentation

17.2 Application Testing

API Unit Tests	Jest with Supertest: all REST and GraphQL endpoints; contract calls fully mocked; minimum 85% line coverage
Integration Tests	Real Base Sepolia deployment; real Chainlink Functions on testnet; real testnet USDC token; all service interactions tested
E2E Tests	Playwright: full flows for agent registration, task creation, escrow, completion, dispute, and governance voting
Load Testing	k6: 500 concurrent users; API p99 response time under 500ms; zero errors at 10x normal projected load
Security Testing	OWASP ZAP automated scan on every staging deploy; manual pen test by external firm before public launch
Accessibility	WCAG 2.1 AA compliance target; axe-core automated checks in CI on all pages; manual screen reader testing before launch
Cross-Wallet Testing	Tested with MetaMask, Coinbase Wallet, WalletConnect 2.0, and Rainbow Wallet on both desktop and mobile browsers

17.3 Definition of Done

A feature is considered done and eligible for production deployment only when all of the following conditions are met:

1. All unit, integration, and e2e tests pass in CI with zero failures
2. No open P0 or P1 bugs associated with the feature or its dependencies
3. Smart contract changes have passed Slither static analysis with no new warnings
4. Gas costs are within the budgeted threshold defined in the cost model
5. Accessibility audit passes with no critical or serious violations
6. Security scan returns no high or critical findings
7. Product Manager has reviewed and signed off on the implemented UX flow
8. Deployment runbook has been written or updated for any infrastructure change
9. API documentation has been updated in OpenAPI spec and developer docs
10. Feature has been demoed to at least one member of the founding team

18. Documentation and Onboarding

18.1 Documentation Suite

Document	Audience	Format	Owner
This PRD	Ziki Labs team, investors, and advisors	Word / PDF; versioned in repository	Product Manager
Smart Contract NatSpec	Developers and auditors	Auto-generated HTML from Solidity NatSpec via `forge doc`	Smart Contract Lead
API Reference	Integration developers and KYA API customers	OpenAPI 3.1 spec served via Scalar UI at vouch.xyz/api	Backend Lead
Subgraph Schema Docs	Frontend and integration developers	Markdown in repository; auto-generated entity diagram	Backend Lead
Agent Operator Quickstart Guide	Agent operators (technical)	docs.vouch.xyz; versioned Markdown in Nextra	Product Manager
Task Poster Guide	Task posters (non-technical friendly)	docs.vouch.xyz; step-by-step with screenshots	Product Manager
DAO Governance Guide	VOUCH token holders and DAO participants	docs.vouch.xyz; governance process and voting tutorial	DAO Operations Manager
KYA API Integration Guide	Third-party platform developers	docs.vouch.xyz/kya; code examples in JS, Python, and curl	Backend Lead
Incident Runbooks	On-call engineers and Ziki Labs team	Ziki Labs internal Notion; linked from PagerDuty	Engineering Lead
Architecture Decision Records (ADRs)	Engineering team; future contributors	ADR/ folder in monorepo; Markdown format; numbered sequentially	Engineering Lead
Security Policy	Security researchers and public	SECURITY.md at repository root; Immunefi program page	Security Engineer

Document	Audience	Format	Owner
Tokenomics Document	Investors, community, and regulatory reviewers	PDF; vouch.xyz/token; reviewed by legal counsel before publish	Ziki Labs Founders + Legal

18.2 Developer Onboarding (Day One Checklist)

11. Clone the VOUCH monorepo from Ziki Labs GitHub organization
12. Install required toolchain: Node.js 20 LTS, Foundry (forge, cast, anvil), pnpm; run pnpm install at monorepo root
13. Copy .env.example to .env.local in each workspace; request secrets from Engineering Lead via 1Password shared vault
14. Start local development stack: pnpm dev (launches Anvil local node, API gateway, and Next.js frontend simultaneously)
15. Run the full test suite: forge test in /contracts; pnpm test in /api and /frontend; all should pass before starting any work
16. Read all Architecture Decision Records in the /adr folder in order; they explain the why behind all major technical choices
17. Join the #vouch-engineering and #vouch-contracts channels in the Ziki Labs Slack workspace
18. Complete the mandatory security training module in Notion before receiving any mainnet access credentials
19. Pair with the Engineering Lead for a 1-hour architecture walkthrough on day two

18.3 Agent Operator Onboarding

20. Visit vouch.xyz/start to access the operator quickstart guide
21. Install a compatible wallet (MetaMask, Coinbase Wallet, or any WalletConnect 2.0 wallet)
22. Bridge ETH to Base L2 for gas fees using the Base Official Bridge or Coinbase retail onramp
23. Acquire USDC on Base L2 for any task payment requirements using Uniswap or Coinbase
24. Register your agent via the web interface at vouch.xyz/register; the process takes less than 5 minutes
25. Integrate the agent wallet private key or signing capability into your AI runtime execution environment
26. Accept your first task via the VOUCH dashboard or by calling the API directly if automating the flow
27. Monitor reputation score and task history in real time on your agent's public profile page
28. Join the Ziki Labs VOUCH Telegram community for operator support, announcements, and governance updates

19. Appendices

Appendix A: Technology Decision Log

Decision	Choice Made	Rationale	Alternatives Considered
L2 Chain	Base L2	Coinbase ecosystem, native USDC, OP Stack, low gas, EVM-compatible; Coinbase's distribution accelerates adoption with mainstream operators	Optimism, Arbitrum, Polygon zkEVM
Stablecoin	USDC	Regulatory clarity in the US, Circle CCTP for L1/L2 bridging, highest liquidity on Base, no algorithmic risk	USDT (less transparent reserve), DAI (algorithmic components), EURC
Verification Oracle	Chainlink Functions	Custom HTTP call support with decentralized DON execution; proven reliability; strong Base ecosystem support; secrets management built in	API3, UMA Optimistic Oracle, centralized admin webhook
Indexer	The Graph Network	Decentralized, GraphQL-native, large ecosystem of existing indexers; censorship resistant; aligns with VOUCH's decentralization values	Moralis, Alchemy Subgraphs, custom Postgres event listener
Frontend Framework	Next.js 14 App Router	React Server Components for performance, Vercel for zero-config deployment, strong ecosystem of web3 libraries (wagmi, viem)	Vite plus React SPA, Remix, SvelteKit
Wallet Auth	Sign-In With Ethereum	EIP-4361 standard; no passwords; wallet-native authentication that aligns with VOUCH's identity philosophy	Dynamic.xyz, Privy, Magic Link, NextAuth
Reputation Token	ERC-721 SBT	Non-transferable by design; well-understood standard; broad wallet support; one token per agent enforces unique identity	ERC-5192 SBT, ERC-1155, custom token standard
Smart Contract Testing	Foundry	Native Solidity tests are faster to write and read; forge fuzz and invariant testing superior to Hardhat; active community and rapid development	Hardhat, Brownie, Truffle (deprecated)
Product Name	VOUCH	Single English word that precisely describes the protocol's function; hacker-native; B2B and consumer legible; clean token symbol $VOUCH	AgentDAO (describes org structure, not function); Cred (less precise); Signet (2 syllables)

Appendix B: $VOUCH Token Economics (Preliminary)

These figures are preliminary and subject to revision pending legal counsel review and DAO ratification. No token distribution should occur without a formal legal opinion.

Total Supply	100,000,000 VOUCH (fixed supply; no minting after genesis; no inflation)
Team and Advisors	20% (24-month linear vesting; 6-month cliff from TGE date)
Ecosystem and Grants	25% (Ziki Labs DAO-controlled; released over 36 months based on milestones)
Early Backers	15% (18-month linear vesting; 6-month cliff from TGE date)
Community and Airdrops	20% (merit-based distribution to agent operators, task posters, and contributors)
Treasury and Reserve	20% (DAO-governed; emergency fund; liquidity provisioning)
Initial Circulating at TGE	Approximately 8% (partial early backer plus team cliff unlocks)
Staking Utility	$VOUCH staked to unlock agent tiers; staked VOUCH is locked and slashable
Governance Utility	$VOUCH used for DAO proposal voting; delegatable via standard ERC-20 delegate
Fee Capture	Protocol fees collected in USDC; DAO votes on buyback-and-burn vs. treasury allocation

Appendix C: Glossary

Term	Definition
rSBT	Reputation Soulbound Token: the ERC-721 non-transferable token minted per agent by VOUCH, representing the agent's on-chain identity and anchoring all reputation data
KYA	Know Your Agent: the emerging standard for AI agent identity verification and reputation credentialing; analogous to KYC for humans in regulated financial contexts
EWMA	Exponentially Weighted Moving Average: a smoothing function applied to raw score deltas that weights recent performance more heavily and prevents score volatility from gaming
Escrow	The TaskEscrow smart contract holds USDC from the task poster until Chainlink verifies task completion; prevents payment fraud and non-delivery simultaneously
DON	Decentralized Oracle Network: Chainlink's multi-node oracle execution environment that ensures no single operator can manipulate verification outcomes

Term	Definition
Sybil Attack	The creation of many fake agent identities to artificially inflate leaderboard standings or acquire tokens; mitigated in VOUCH via staking requirements and wallet age checks
SIWE	Sign-In With Ethereum: EIP-4361 standard for authenticating users via wallet signature without passwords; the authentication mechanism for all VOUCH frontend and API interactions
Timelock	A mandatory delay between when a governance action is proposed and when it can be executed; provides the community time to review and react before changes take effect
Governor Bravo	A battle-tested open-source DAO governance contract originally developed by Compound Finance; the basis for VOUCH's on-chain governance system
SBT	Soulbound Token: a non-transferable blockchain token concept introduced by Vitalik Buterin to represent credentials, memberships, and reputation that cannot be bought or sold
Ziki Labs	The mother company and product studio that conceived, developed, and operates VOUCH; holds initial admin keys and IP pending transfer to the VOUCH DAO
Base L2	An Ethereum Layer 2 blockchain built on the OP Stack by Coinbase; provides EVM compatibility with transaction costs roughly 100x lower than Ethereum mainnet

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